

Nurture COURSE

PHASE : SPEED & ACCURACY TEST

TARGET : PRE-MEDICAL 2023

Test Type : **PRACTICE TEST**

Test Pattern : **NEET (UG)**

TEST DATE : _____

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	4	1	1	2	4	3	3	3	4	2	1	3	2	3	4	1	3	3	1	2	4	4	4	1	1	2	1	3	3	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45															
A.	4	1	1	4	2	2	4	1	2	1	4	1	4	1	3															



HINT - SHEET

1. **Ans (4)**

The frequency is a characteristic of source. It is independent of the medium. Hence, the correct option is 600 Hz.

2. **Ans (1)**

For the end correction e,

$$\Rightarrow e = \frac{l_2 - 3l_1}{2} = \frac{70.2 - 3 \times 22.7}{2} = 1.05 \text{ cm}$$

3. **Ans (1)**

$$\text{Intensity} \propto (\text{amplitude})^2 \Rightarrow I \propto (2a \cos kx)^2$$

Hence intensity will be maximum when $\cos(kx)$ is maximum

4. **Ans (2)**

As the sound waves are mechanical waves they requires medium for propagation.

5. **Ans (4)**

$$\text{Speed of wave} = v = \sqrt{\frac{T}{\mu}}$$

$$\Rightarrow v = \frac{\omega}{k} = \frac{0.50}{0.04} = \sqrt{\frac{T}{0.04}}$$

$$T = 6.25 \text{ N}$$

6. **Ans (3)**

Sonometer is based on Resonance.

7. **Ans (3)**

As there is no effect of change in pressure on velocity of sound in air.

$$\text{Speed of sound} = v = \sqrt{\frac{\gamma RT}{M}} \Rightarrow v \propto \sqrt{T}$$

$$\frac{v_2}{v_1} = \sqrt{\frac{2T}{T}} \Rightarrow v_2 = 330\sqrt{2} \text{ m/s}$$

8. **Ans (3)**

$$\text{Energy (E)} \propto A^2 f^2$$

Here Amplitude (A) is the same in both the cases, but the frequency in the second case is two times the frequency in first case, Therefore $E_2 = 4E_1$

9. **Ans (4)**

Energy density of wave is given by

$$u = 2\pi^2 n^2 \rho a^2$$

Or $u \propto a^2$ (as n and ρ are constant)

$$\frac{u_1}{u_2} = \frac{a_1^2}{a_2^2} = \frac{5}{2} = \frac{25}{4}$$

10. **Ans (2)**

$$l_1 = 18 \text{ cm } f = \frac{v_1}{4l_1} \text{ (for first resonance) and}$$

$$f = \frac{3v_2}{4l_2} \text{ (for second resonance),}$$

$$\text{Where } l_2 = x$$

According to given situation

$$\frac{3v_2}{4l_2} = \frac{v_1}{4l_1} \text{ (or) } l_2 = 3l_1 \times \frac{v_2}{v_1}; v = \sqrt{\frac{\gamma RT}{M}}$$

$$X = 54 \times (\text{A quantity greater than 1})$$

Also $v_1 < v_2$ as during summer temperature would be higher. So $X > 54$

11. **Ans (1)**

$$y_1 = 10^{-6} \sin \left[100t + \left(\frac{x}{50} \right) + 0.5 \right]$$

$$y_2 = 10^{-6} \cos \left[100t + \left(\frac{x}{50} \right) \right]$$

$$= 10^{-6} \sin \left[100t + \left(\frac{x}{50} \right) + \frac{\pi}{2} \right]$$

$$= 10^{-6} \sin \left[100t + \left(\frac{x}{50} \right) + 1.57 \right]$$

$$[\text{Using } \cos x = \sin(x + \frac{\pi}{2})]$$

$$\text{The phase difference} = 1.57 - 0.5 = 1.07$$

$$[\text{Or using } \sin x = \cos(\frac{\pi}{2} - x)]$$

We get the same result.

12. **Ans (3)**

$$X_1 = 12 \sin(484\pi t - 7\pi x)$$

$$= 12 \sin 2\pi \left(242t - \frac{7x}{2} \right)$$

$$\therefore \text{Frequency } n_1 = 242 \text{ Hz}$$

$$X_2 = 12 \sin(480\pi t - 7\pi x)$$

$$= 12 \sin 2\pi \left(240t - \frac{7x}{2} \right)$$

$$\therefore \text{Frequency } n_2 = 240 \text{ Hz}$$

$$\text{Beat frequency} = n_1 - n_2 = 2$$

13. **Ans (2)**

Speed of sound in water is given by,

$$v = \sqrt{\frac{B}{\rho}} = \sqrt{\frac{2100 \times 10^6}{1000}} = \sqrt{2.1 \times 10^6}$$

$$\text{or } v = 1.45 \times 10^3 = 1450 \text{ m/s}$$

14. **Ans (3)**

Given frequency $f = 500 \text{ Hz}$,

velocity $v = 350 \text{ m/s}$,

phase difference $(\Delta\phi) = 60^\circ$

$$\text{By the formula, } \lambda = \frac{v}{f} = \frac{350}{500} = 0.7 \text{ m}$$

$$\text{Also we know that } \Delta x = \Delta\phi \frac{\lambda}{2\pi}$$

$$= 60 \times \frac{\pi}{180} \times \frac{0.7}{2\pi} \approx 0.12 \text{ m} \approx 12 \text{ cm}$$

15. **Ans (4)**

As man is moving towards the cliff, so apparent frequency heard directly towards source will be

$$f_2 = \frac{330 - v}{330 - 0} \times 600$$

Similarly, apparent frequency heard by observer after reflection from the cliff will be

$$f_3 = \frac{330 + v}{330 - 0} \times 600$$

Now these two frequency f_2 and f_3 will produce 10 beats per second therefore

$$f_3 - f_2 = 10 \text{ by solving we get } v = 2.75 \text{ m/s}$$

16. **Ans (1)**

Suppose n_A is known frequency, then $n_A = 100 \text{ Hz}$, and let n_B be the frequency of the other tuning fork

$X = 2 =$ beat frequency, which is decreasing after loading (i.e. $x \downarrow$)

Unknown tuning fork is loaded so n_B decreases and also given that beat frequency is 1, so beat frequency also decreases

$$\text{Hence, } n_B > n_A \text{ and } n_B - n_A = x$$

$$n_B = n_A + x = 100 + 2 = 102 \text{ Hz}$$

17. **Ans (3)**

Comparing with the standard equation,

$y = A \sin(kx - \omega t)$ we have

$$\omega = 2\pi \text{rad/s}$$

$$\text{Frequency } n = \frac{\omega}{2\pi} = \frac{2\pi}{2\pi} = 1 \text{ s}^{-1}, n = 1 \text{ s}^{-1}$$

18. **Ans (3)**

Let d be the distance of epicentre

Time taken by S – wave

$$= \frac{d}{6}, \text{ Time taken by P – wave} = \frac{d}{9}$$

$$\text{Now, } 5 \times 60 = \left(\frac{d}{6} - \frac{d}{9} \right)$$

$$\Rightarrow 3d = 300 \times 54, d = 5400 \text{ km}$$

19. **Ans (1)**

According to the Doppler's effect, whenever there is relative motion between a source of sound and listener, the apparent frequency of sound heard by the listener is different from the source. Let S be source of sound. Let v be the actual frequency of sound emitted by the source and λ be the actual wavelength of the sound emitted.

If v is velocity of sound in still air, then $\lambda = \frac{v}{V}$

If velocity of listener is v_L and velocity of source is v_S , then apparent frequency of sound waves heard by the listener is

$$v' = \frac{v - v_L}{v - v_S} \times V$$

Here both source and listener are approaching each other. Then v_S is positive and v_L is negative.

$$\text{So, } v' = \frac{v - (-v_L)}{v - v_S} \times V = \left\{ \frac{v + v_L}{v - v_S} \right\} V$$

$$\rightarrow v' > v \text{ also, } \lambda' < \lambda$$

So, listener listen more frequency and observes less wavelength.

20. **Ans (2)**

Frequency in transverse wave

$$n = \frac{1}{2l} \sqrt{\frac{T}{\mu}} \Rightarrow \frac{n_2}{n_1} = \sqrt{\frac{T_2}{T_1}} = \sqrt{\frac{81}{100}} = \frac{9}{10}$$

$$\therefore \left(\frac{n_1 - n_2}{n_1} \right) \times 100 = 10\%$$

21. **Ans (4)**

The intensity can be given by

$$I = 2\pi^2 \rho^2 A^2 v$$

Hence

$$\frac{I_A}{I_B} = \frac{f_A^2 A_A^2}{f_B^2 A_B^2}$$

$$\frac{I_A}{I_B} = \left(\frac{1}{2} \right)^2 \left(\frac{2}{1} \right)^2$$

$$\frac{I_A}{I_B} = 1$$

22. **Ans (4)**

$$I_{\max} \propto A_{\max}^2$$

$\Downarrow$

$$I_{\max} \propto (a + a)^2$$

$\Downarrow$

$$I_{\max} = \text{const. } (4a^2)$$

23. **Ans (4)**

The equation can be written as

$$y = 4 \sin \left(4\pi t - \frac{\pi x}{16} \right)$$

Wave velocity is given by

$$v = \frac{\text{coefficient of } t (\omega)}{\text{coefficient of } x (k)}$$

$$\Rightarrow v = \frac{4\pi}{\frac{\pi}{16}} = 64 \frac{\text{cm}}{\text{s}} \text{ in } + \text{vex - direction.}$$

24. **Ans (1)**

Phase difference is means constructive interference so resultant amplitude will be maximum.

25. **Ans (1)**

Simple harmonic waves are setup in the string fixed at both two ends. Earth spinning about its own axis, ball bouncing between two rigid vertical walls and particle moving in a circle with uniform speed, all these are periodic but not simple harmonic

26. **Ans (2)**

$$\Rightarrow f_{app} = f_0 \left(\frac{v + v_0}{v} \right)$$

$$\Rightarrow \frac{f_{app} - f_0}{f_0} = \frac{v_0}{v} = \frac{v}{5} = \frac{1}{5} = 20\%$$

27. **Ans (1)**

$$V_{max} = a\omega$$

$$V_{max} = a \times 2\pi f$$

$$V_{max} = 0.12\pi \times 300$$

$$V_{max} = 60\pi \text{ cm s}^{-1}$$

28. **Ans (3)**

The time interval between two consecutive beats

$$T = \frac{1}{n_1 - n_2} = \frac{1}{260 - 256} = \frac{1}{4} \text{ s}$$

$$\text{So, } t = \frac{1}{16} = \frac{T}{4} \text{ s}$$

By using,

$$\frac{\Delta\phi}{2\pi} = \frac{\Delta T}{T}$$

$$\Delta\phi = \frac{2\pi(\Delta T)}{T} = \frac{2\pi \times T/4}{T} = \frac{\pi}{2}$$

29. **Ans (3)**

Phase difference = $2\pi/\lambda \times$ path difference

$$\Rightarrow 1.6\pi = \frac{2\pi}{\lambda} \times 40$$

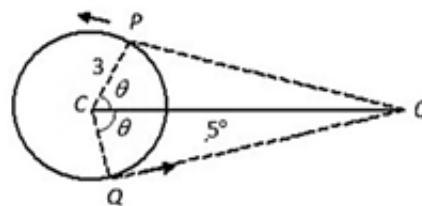
$$\Rightarrow \lambda = 50 \text{ cm} = 0.5 \text{ m}$$

$$\Rightarrow v = n\lambda$$

$$\Rightarrow 330 = 0.5 \times n$$

$$\Rightarrow n = 660 \text{ Hz}$$

30. **Ans (3)**



When the source is at point P, the frequency received is minimum and when the source is at point Q, the frequency received is maximum.

$$2\theta = \omega t \Rightarrow t = \frac{2}{5} \cos^{-1} \left(\frac{3}{5} \right)$$

31. **Ans (4)**

$y = A \sin^2(kx - \omega t + \phi)$ can be rewritten as

$$y = \frac{A}{2} - \frac{A}{2} \cos(2kx - 2\omega t + 2\phi)$$

Therefore the amplitude of the wave is $A/2$ and wavelength of wave is π/k

32. **Ans (1)**

The speed of transverse wave,

$$v = \sqrt{\frac{T}{m}} = \sqrt{\frac{T}{\mu}}$$

$$\therefore v = \sqrt{\frac{20 \times 10^{-3}}{7.5 \times 0.2 \times (10^{-3})^2}}$$

$$v \approx 116 \text{ ms}^{-1}$$

33. **Ans (1)**

$$l_1 = 0.516 \text{ m}, l_2 = 0.491 \text{ m}, T = 20 \text{ N}$$

Mass per unit length, $\mu = 0.001 \text{ kg/m}$

Frequency,

$$v = \frac{1}{2l} \sqrt{\frac{T}{\mu}}$$

$$v_1 = \frac{1}{2 \times 0.516} \sqrt{\frac{20}{0.001}}$$

$$v_2 = \frac{1}{2 \times 0.491} \sqrt{\frac{20}{0.001}}$$

So, number of beats = $v_1 - v_2 = 7$

34. **Ans (4)**

$$\frac{n}{n} = \left(\frac{V}{V - V_s} \right)$$

$$\frac{10000}{9500} = \frac{300}{300 - V_s}$$

V_s = speed source

$$(300 - V) = 285 \Rightarrow V = 15 \text{ m/s}$$

35. **Ans (2)**

$$\text{Since, } \frac{v}{c} = \frac{\Delta\lambda}{\lambda}$$

$$\therefore \Delta\lambda = \frac{v}{c} \times \lambda$$

$$\text{or } \frac{\Delta\lambda}{\lambda} = \frac{1.2 \times 10^6}{3 \times 10^8} = \frac{12}{3} \times \frac{10^5}{10^8} = 4 \times 10^{-3}$$

$$\frac{\Delta\lambda}{\lambda} \times 100 = 0.4 \text{ i.e. } 0.4\%$$

36. **Ans (2)**

Average intensity along the string

$$\langle I \rangle = 2\pi^2 f^2 A^2 \rho v$$

$$\frac{I_1}{I_2} = \left(\frac{f_1 A_1}{f_2 A_2} \right)^2 = \left(\frac{T_2 A_1}{T_1 A_2} \right)^2 = \left(\frac{1}{2} \times 3 \right)^2 = \frac{9}{4}$$

37. **Ans (4)**

$$L_1 = 10 \log_{10} \left(\frac{I_1}{I_0} \right)$$

$$L_2 = 10 \log_{10} \left(\frac{I_2}{I_0} \right)$$

$$L_2 - L_1 = DL = 10 \log_{10} \left(\frac{I_2}{I_1} \right)$$

$$= 10 \log_{10} \left(\frac{P_2}{P_1} \right) \quad (I \propto \text{Power})$$

$$= 10 \log_{10} \left(\frac{400}{10} \right)$$

$$= 10 \log_{10}(40)$$

$$= 10 [\log_{10}(4) + \log_{10}(10)]$$

$$= 10[2(0.3010) + 1]$$

$$\cong 16 \text{ db}$$

38. **Ans (1)**

$$\text{Fundamental frequency } f = \frac{1}{2l} \sqrt{\frac{T}{\mu}}$$

$$\frac{f_1}{f_2} = \frac{l_2}{l_1} \sqrt{\frac{\mu_2}{\mu_1}} \text{ (since tension is same)}$$

$$= \frac{2L}{L} \sqrt{\frac{\pi r p}{\pi 4 r p}} \text{ (since wires are of the same material)}$$

Hence the desired ratio is 1

39. **Ans (2)**

Let the frequency of standard for $k = x$

$$n_A = \frac{102}{100}x, n_B = \frac{97}{100}x$$

$$\text{Number of beats per second} = n_A - n_B = 6$$

$$\frac{102}{100}x - \frac{97}{100}x = 6$$

$$x = \frac{6 \times 100}{5} = 120 \text{ Hz}$$

$$n_A = \frac{102}{100}x = 122.4 \text{ Hz}$$

40. **Ans (1)**

Frequency of second overtone (fifth harmonic)

$$\text{of close pipe} = \frac{5v}{4l}$$

Frequency of first overtone (second harmonic)

of an open pipe $= 2v/2l$, Accordingly,

$$\frac{5v}{4l} - \frac{2v}{2l} = 100 \text{ or}$$

$$\frac{v}{4l} = 100 \text{ or } v = 400l$$

The fundamental frequency of open pipe

$$= \frac{v}{2l} = \frac{400l}{2l} = 200 \text{ S}^{-1}$$

41. **Ans (4)**

From the equation, $k = 0.157$

$$\frac{2\pi}{\lambda} = 0.157.$$

$$\lambda = 40 \text{ m}$$

But the length of the wire in 4th harmonic,

$$l = 4 \times \frac{\lambda}{2} = 2\lambda = 80 \text{ m}$$

42. **Ans (1)**

The average power per unit area that is incident perpendicular to the direction of propagation is called the intensity. Intensity of sound

$$I = \frac{P}{4\pi r^2} \Rightarrow I \propto \frac{1}{r^2} \Rightarrow \frac{I_1}{I_2} = \left(\frac{r_2}{r_1} \right)^2$$

Here $r_1 = 2\text{m}$, $r_2 = 3\text{m}$

Substituting the values, we have

$$\frac{I_1}{I_2} = \left(\frac{3}{2} \right)^2 = \frac{9}{4}$$

43. **Ans (4)**

$$y = 5(\sin 4\pi t + \sqrt{3}\cos 4\pi t)$$

$$y = (5\sin 4\pi t + 5\sqrt{3}\cos 4\pi t)$$

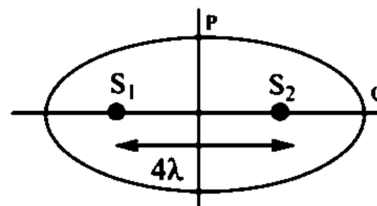
$$A = \sqrt{A_1^2 + A_2^2}$$

$$A = \sqrt{(5)^2 + (5\sqrt{3})^2}$$

$$= \sqrt{25 + 75} = \sqrt{100}$$

$$A = 10$$

44. **Ans (1)**



$$S_1P - S_2P = 0$$

$$S_1Q - S_2Q = 4\lambda$$

3 maximum lie on are between P and Q.

Counting all quadrants; the total number is 12.

P and Q symmetrical points are also maxima.

Total number is 16

45. **Ans (3)**

The explanation of the statements are given below

(i) In Meld's experiment $p \cdot \sqrt{T} = \text{constant}$. $p^2 T = \text{constant}$. Hence, this statement is correct.

(ii) In Kundt's experiment distance between two heaps of powder is $\lambda/2$, Hence, this statement is correct

(iii) Quink's tube experiment is related to interference, hence the option is incorrect

(iv) Echo phenomena are related to the reflection of sound. So, this statement is correct.